

Chapter 13: Geometry of Vectors

Recall: For two vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ in $\mathbb{R}^n$, the **dot product** is

$$\mathbf{x} \cdot \mathbf{y} =$$

Definition: Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be a vector in $\mathbb{R}^n$. Then the **length**, or **magnitude**, or **norm** of the vector $\mathbf{x}$ is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$$

In particular, for $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in $\mathbb{R}^2$, the magnitude of $\mathbf{v}$ is

$$\|\mathbf{v}\| =$$

Example: What is the length/magnitude of $\mathbf{v} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$?

Example: What is the length/magnitude of the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$?

Example: Compute the distance between the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}.$$

Fact: In general, the distance between two vectors $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ in $\mathbb{R}^n$

is $\|\mathbf{v} - \mathbf{u}\|$, which can be written out as

$$\|\mathbf{v} - \mathbf{u}\| =$$

Theorem: Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in $\mathbb{R}^n$ and let a be a scalar. Then the following properties for dot product hold:

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$
2. $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$
3. $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$
4. $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$
5. $\mathbf{x} \cdot \mathbf{x} \geq 0$ and $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$
6. $\|a\mathbf{x}\| = |a| \|\mathbf{x}\|$

Proof of _____ (Pick one):

(more space for proof):

Theorem: (Cauchy–Bunyakovsky¹–Schwarz inequality)

For every pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, the following inequality holds:

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Proof: We'll do this proof later in more generality (like Bunyakovsky and Schwarz).

If $\mathbf{u}, \mathbf{v}$ are nonzero vectors, we can rewrite the CBS inequality as:

This allows us to define the following:

Definition: The **angle** θ between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ obeys the formula

$$\cos \theta =$$

Example: Let $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. What is the angle θ between $\mathbf{u}$ and $\mathbf{v}$?

¹Most textbooks (including yours) Call this the Cauchy-Schwarz inequality. Cauchy proved the most basic version of this theorem, while Bunyakovsky and Schwarz independently proved more general versions of this theorem.

Theorem: (Triangle Inequality) For every pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, we have $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Let's draw a picture to see why this makes sense in $\mathbb{R}^2$:

Proof: (If time):

Question: When will $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$?

Definition: A **unit vector** in $\mathbb{R}^n$ is a vector whose length/magnitude is one.

Fact: If $\mathbf{x}$ is any nonzero vector, then the vector

$$\mathbf{u} =$$

is a unit vector in the direction of $\mathbf{x}$.

Example: Let $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix}$. Find a unit vector $\mathbf{u}$ in the direction of $\mathbf{x}$.

Example: Consider the set of all unit vectors in $\mathbb{R}^2$. Visually, what do these create?

Chapter 14: Orthogonal Vectors

Example: Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -6 \\ 0 \\ 4 \end{bmatrix}$. What is the angle θ between $\mathbf{u}$ and $\mathbf{v}$?

Definition: Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal**, or **perpendicular**, if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

Example: Which pair(s) of the following three vectors are perpendicular?

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -2 \\ -6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

Example: Consider $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$. Find the set of all vectors which are orthogonal to $\mathbf{v}$ in $\mathbb{R}^3$.

Orthogonal Projection

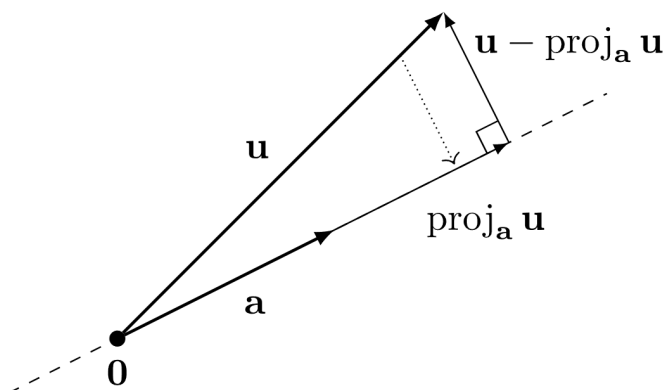
Consider two vectors $\mathbf{u} = \overrightarrow{OQ}$ and $\mathbf{a} = \overrightarrow{OR}$, which both have initial point at the origin, and terminal points at Q and R respectively. Consider the line ℓ through $\mathbf{a}$.

Question: What is the shortest distance from the terminal point Q to the line ℓ ?
put tikz here.

Definition: The **orthogonal projection** of $\mathbf{u}$ onto $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$$

When the initial point of $\text{proj}_{\mathbf{a}} \mathbf{u}$ is placed at the origin, the terminal point will be the point closest to $\mathbf{u}$ through the origin and parallel to $\mathbf{a}$.



Fact: Vectors $\mathbf{a}$ and $\mathbf{u} - \text{proj}_{\mathbf{a}} \mathbf{u}$ are orthogonal.

Definition: The vector $\mathbf{u} - \text{proj}_{\mathbf{a}} \mathbf{u}$ is called the **vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$** .

Note: Your homework calls this the “orthogonal complement.”

Example: Find the orthogonal projection of $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ onto $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and shortest distance from $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ to the line containing $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$.